Contextual social navigation through integrated task and motion planning

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Abstract. In real-world scenarios, robots often interact with humans across various levels of abstraction, posing research challenges like task allocation, human-robot joint actions, and social navigation. To achieve seamless interactions and make robots more acceptable, they must adapt their behavior based on context, e.g., to navigate social contexts effectively. This paper proposes an approach integrating task and motion planning to contextualize robot behaviors for social navigation. It leverages the contextual knowledge of a deliberative task planner to adjust the robot navigation dynamically, improving human-robot interaction. The proposed framework can handle varied sequences of social navigation tasks, and adapt navigation behaviors to the interacting features of involved human users. We validate this approach through a simulated hospital scenario and show an in-laboratory deployment on a real robot.

Keywords: Social Navigation · Task and Motion Planning · Socially Assistive Robotics.

1 Introduction

Robots acting in real-world situations usually require direct or indirect interactions with humans to realize behaviors that consider the social dimension. Robot behaviors should be technically efficient and acceptable to humans [20]. In Human-Robot Interaction (HRI), the social perspective is crucial when dealing with humans to meet their expectations and realize safe, reliable, and legible behaviors. It is important to reason about *how* tasks are carried out by a robot to do the *right action* in the *right way* and comply with the so-called *social norms* [24, 4, 1]. Furthermore, the presence of humans constitutes a source of *uncertainty* concerning, e.g., their goals, beliefs, and intentions [8, 9]. This uncertainty raises robot control issues and strongly affects the design of robot controllers [10, 16]. To deploy more natural and acceptable behaviors, robots need advanced *reasoning capabilities* that take into account: *who* is the human a robot interacts with; *what* are the objectives of the interactions; *how* to achieve them; *when* to execute the needed actions, and; *where* interactions take place. Integrated Robotics and Artificial Intelligence (AI) would enhance robot autonomy and synthesize adaptive behaviors by reasoning technical and social knowledge [14, 12].

In this work, we propose an integrated Task And Motion Planning (TAMP) approach to enhance the awareness of the social navigation skills of robots. This approach relies on a motion planning framework, called CoHAN [22, 23], which allows the tuning of navigation behaviors by pushing human awareness. It exposes several motion parameters the task planner PLATINUm [26] uses to adapt navigation behaviors to the changing social context. To this aim, the paper proposes an integrated holistic model characterizing task requirements and human features at both domain and geometric levels. We design an assistive scenario requiring the robot to implement interacting services under different technical and social conditions. Each service requires the robot to autonomously adapt navigation behaviors to social contexts, task priorities, and human *qualities*. We assess the TAMP approach on this scenario to show the improved *social autonomy*.

2 Human-Aware Task & Motion Planning

Well-structured models of humans are crucial for realizing robot behaviors that are effective and acceptable in social contexts. Several research efforts have focused on explicitly modeling human factors to enhance the social awareness of robot behaviors [2, 17]. These models are integrated into the geometric model of the robot to plan trajectories and find a balance between efficiency and social acceptance of robot motions [21, 13]. Humans are for example considered dynamic obstacles, with geometric models accounting for factors such as proxemics [11, 19, 15] or emotional states [5]. Additional factors like human intentions, perspectives, and social norms have been also considered [6, 18, 3]. Although researchers have successfully integrated human factors into the geometric model of navigation controllers, geometric-level planning decisions usually focus on single navigation tasks or interaction episodes. These decisions typically evaluate the best trajectories to the next navigation goal, ignoring more abstract knowledge about human interaction skills, sequences of navigation goals, and the motivations behind the robot's actions.

We propose a human-aware control approach integrating task and motion planning (TAMP) to characterize navigation behaviors from different synergetic perspectives: (i) a *domain perspective* considers the technical and performance aspects of the task being executed; (ii) a *human perspective* considers the qualities of the humans involved in a task; (iii) an *environment perspective* considers the social features of the environment. The architecture is implemented in ROS Melodic using: (i) PLATINUm [4] [26] as a timeline-based task planner; (ii) ROXANNE [5] [25] as ROS-compliant executive for timeline-based plans, and; (iii) CoHAN [6] [22] as a motion planner implementing the navigation skills.

Fig. 1 shows the proposed approach. The control flow of the robot is initiated by the task planner which receives planning requests through the goal topic of

[4] `https://github.com/pstlab/PLATINUm.git`
[5] `https://github.com/pstlab/roxanne_rosjava.git`
[6] `https://github.com/sphanit/cohan_planner_multi`

ROXANNE. For each request, PLATINUm synthesizes a plan by decomposing the requested task into temporal flexible (motion) actions that are then executed by ROXANNE. Navigation actions are dispatched to CoHAN which plans geometric trajectories to implement robot motions.

CoHAN exposes to the task planner a set of navigation parameters suitable to tailor trajectory planning to the different social contexts within a domain-level task. These parameters are grouped into robot, human, and social sets. Robot variables determine the qualities of robot motions. They control the speed and acceleration bounds of the robot and the *look ahead* of the trajectory to adapt the planning horizon and environmental reactivity. Human variables determine the qualities of the expected humans in a "navigation episode". They describe the velocity and acceleration of the human. Furthermore, they include *radius* and *field of vision* variables that specify respectively proxemics constraints for robot motions and the breadth of the field of vision of the human to evaluate the visibility of the robot from the human's point of view. Finally, social variables determine additional constraints to consider *social acceptance* of planned motions. The parameter *visibility* determines the minimum distance from the human when the robot enters the human's field of vision. Similarly, the parameter *hidden human* allows the robot to be cautious about occluded regions where a human might emerge unexpectedly.

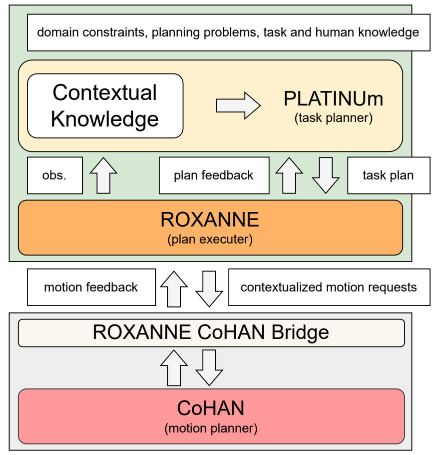


Fig. 1: Integrated Task and Motion planning architecture.

The task planner reasons at the domain level by considering domain requirements, functional capabilities of the robot, and human interacting qualities. PLATINUm uses such navigation parameters to dynamically constrain possible navigation behaviors according to the different features characterizing the execution of complex (domain-level) tasks.

Contribution: The rationale behind the proposed TAMP approach is to leverage task-level knowledge about priorities, physical and cognitive features of involved humans, environmental structure to *contextually* set navigation parameters. The resulting integrated control flow would enhance the *social autonomy* of the robot by tuning motion behaviors to the various social situations encountered during the execution of a task. The central contribution of the work thus consists of the mapping of (task-level) contextual knowledge about the expected social interactions and the (geometric-level) navigation skills of the robot. The resulting goal-oriented autonomy enhances the *awareness* of a robot. A robot would indeed autonomously evaluate the technical and social consequences of a requested task and accordingly set the most suitable navigation (or interaction) parameters.

3 An Assistive Domain

We consider a social robot carrying out assistive tasks inside a hospital. The choice aims to emphasize the need for contextual and social awareness as behavioral qualities necessary for the robot to act effectively. Assistive domains are indeed characterized by various social situations that may affect robot behaviors in different ways. On the one hand, tasks may have different priorities and technical requirements. On the other hand, the execution of the same task may require interacting with different humans (e.g., patients, doctors) and environments (e.g., patients' rooms, corridors, technical rooms).

The scenario consists of three assistive tasks. (i) A *drug delivery task* requires the robot to reach the pharmacy, pick up some drugs, and deliver them to a particular patient located in a known room; (ii) A *patrolling task* requires the robot to visit the rooms of the floor and monitor the general health conditions of patients.; (iii) An *emergency task* requires the robot to rush to the room hosting the patient asking for help. All tasks require the robot to navigate different areas (e.g., rooms or corridors), and interact with different humans (e.g., patients or healthcare professionals) under heterogeneous conditions (e.g., high-priority critical tasks or low-priority nominal tasks).

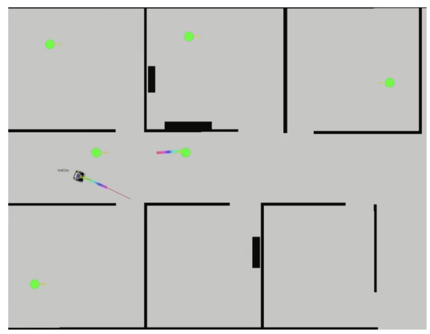


Fig. 2: Layout of the environment.

Fig. 2 shows a schema of the designed hospital map and the distribution of the humans. Six humans are considered in the setting. Three patients are placed in their respective rooms (upper part of the map). Three clinicians are located in the corridor (two) and the pharmacy (the room in the bottom left corner of the map). It is worth noticing that the designed scenario requires the robot to change motion behaviors within a single domain-level task. Namely, the execution of each task can be split into sequential "navigation episodes" each entailing specific contextual knowledge. It is therefore central the capability of adapting motions to the specific requirements of each task, area, and human.

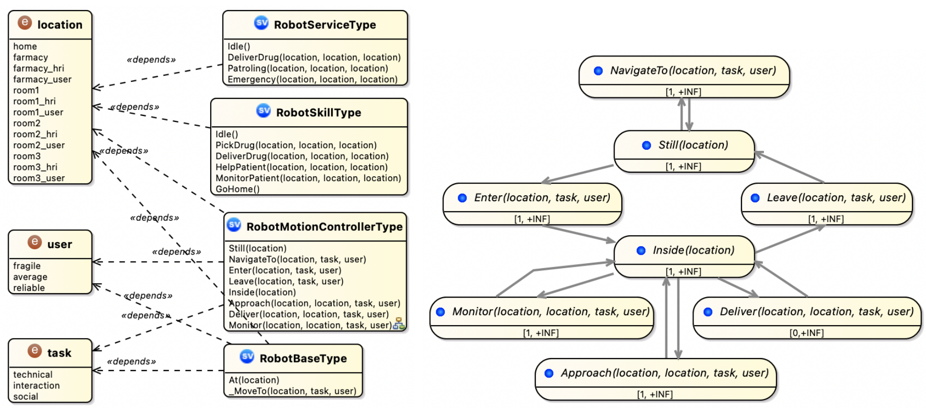
3.1 Task Planning Model

The task planning model describes the "reasoning rules" that allow the robot to achieve the desired objectives. We model the assistive tasks by following the timeline-based formalism [7][7]. The model consists of four state variables whose values denote predicates asserting states or activities a particular domain feature may assume or execute over time. The state variable *RobotBase* models the wheeled base allowing the robot to navigate the environment. The value *At(?l)*

[7] Due to space limitations, readers may refer to [7] for an exhaustive description of planning formalism.

specifies the robot is still in a specific location (i.e., the parameter *location* of the predicate). The value *MoveTo(?l, ?t, ?u)* models the generic action of moving the robot towards a particular goal location (?l). The other parameters contextualize the motion according to the class of task (?t) and the class of assisted human (?u). The value *MoveTo(?l, ?t, ?u)* is modeled as *partially controllable* since humans may affect its execution [7].

The state variable *RobotMotionController* abstracts navigation skills by defining contextual motion primitives. The values *Still(?l)* and *NavigateTo(?l, ?t, ?u)* characterize motions outside the rooms. The values *Enter(?l, ?t, ?u)*, *Inside(?l)*, *Approach(?l, ?t, ?u)*, *Leave(?l, ?t, ?u)* characterize motions inside the rooms. The state variable *RobotSkill* abstracts the interacting skills. The values represent complex actions (e.g., *PickDrug(?l1, ?l2, ?l3)*, *DeliverDrug(?l1, ?l2, ?l3)*) the robot should perform to carry out tasks. In turn, the state variable *RobotService* describes the high-level tasks a robot is capable of performing i.e., *DeliverDrug(?l1, ?l2, ?l3)*, *Patrolling(?l1, ?l2, ?l3)*, and *Emergency(?l1, ?l2, ?l3)*.



(a) (b)

Fig. 3: Task planning model: (a) set of state variables and symbols; (b) detail of the *RobbotMotionController* state variable.

Fig. 3(a) summarizes the modeled state variables. In general, they have a simple structure with a single value as a stable state (e.g., the values *Idle()*, *Still(?l)* and *At(?l)*) and other values as actions with bounded duration (e.g., *DeliverDrug(?l1, ?l2, ?l3)*, *PickDrug(?l1, ?l2, ?l3)*, *HelpPatient(?l1, ?l2, ?l3)*). Transitions are allowed between the stable state and action values (e.g., *Idle()* to action *DeliverDrug(?l1, ?l2, ?l3)* and back). The state variable *RobotMotionController* is more complex. As shown in Fig. 3(b), it distinguishes between the robot moving inside the corridor (i.e., *NavigateTo(?l, ?t, ?u)*) and inside rooms (e.g., *Enter(?l, ?t, ?u)*, *Approach(?l, ?t, ?u)*) to fine-tune navigation parameters.

Synchronization rules hierarchically decompose higher-level tasks (i.e., values belonging to more abstract state variables) into lower-level tasks (i.e., values belonging to lower-level state variables). For example, the task *DeliverDrug(?l1, ?l2, ?l3)* is decomposed into values of the state variable *RobotSkill* specifying complex actions necessary to accomplish the task correctly (e.g., *PickDrug(?l1,*

?l2, ?l3)). Each complex action is further decomposed into contextualized navigation skills (e.g., *NavigateTo(?l, ?t, ?u)*). In addition to causal decomposition, rules specify temporal constraints between decomposed tasks to constrain their execution. For example, the decomposition of the high-level task *DeliverDrug(?l1, ?l2, ?l3)* specifies precedence constraints between the entailed simpler tasks *PickDrug(?l1, ?l2, ?l3)* before *DeliverDrug(?l1, ?l2, ?l3)*.

The (task) planning process decomposes assistive tasks into robot motions by instantiating navigation predicates (i.e., grounding of *MoveTo(?l, ?t, ?u)* predicates). Grounded primitive predicates (i.e., instances of no-further decomposable values) are dispatched to the motion planner for execution. The grounded parameters of dispatched motions encapsulate the contextual knowledge constraining the trajectory planning of the motion planner.

3.2 Combining Symbolic and Geometric Planning

Each time a robot performs a task, planning decisions should balance the technical and social aspects of the resulting behaviors. Highly efficient motions could be risky or not acceptable to humans. Conversely, highly constrained motions might not meet the required deadline (i.e., performance). Task-level knowledge should characterize the motivations and objectives that lead a robot to act in the (social) environment.

We define three classes of assistive tasks: (i) Technical; (ii) Interaction, and; (iii) Social. Intuitively, technical tasks should focus on the technical requirements mainly. Social tasks should focus on social requirements. Interaction tasks are in the middle and balance technical and social requirements. In addition, we define three classes of human actors (i) Fragile; (ii) Average, and; (iii) Reliable. Fragile humans are highly unpredictable and require conservative/prudent behaviors. Reliable humans are less uncertain and make some room for optimization. Average humans require a trade-off between conservative and optimized motions.

Table 1: Mapping Task classes to robot motion parameters.

Task Class	Velocity	Angular Velocity	Acceleration	Planning Horizon	Band Tightness
Technical	max	max	max	min	tight
Interaction	nominal	nominal	nominal	nominal	medium
Social	min	min	min	max	loose

Depending on the classification of the tasks and the humans, the task planner determines *patterns* of motion parameter values to constrain trajectory planning and enforce context awareness. Table 1 and Table 2 map the three classes of tasks and the three classes of humans respectively to the motion variables of the robot and the human.

Table 2: Mapping Human classes to human motion parameters.

Human Class	Radius	Velocity	Angular Velocity	Field of Vision	Band Tightness
Fragile	big	min	min	narrow	tight
Average	medium	nominal	nominal	nominal	medium
Reliable	small	max	max	wide	loose

Table 3 maps combinations of task/human classes to the social motion variables. Following this mapping, for example, *social-critical tasks* requiring the interaction between a robot and a *fragile user* entail a prudent navigation behavior of the robot. The task planner would thus dispatch navigation requests by setting the variable *Radius* to "big" (i.e., avoid the robot moving too close to the human); the variable *Field of vision* to "narrow" (i.e., assume the user would see the robot only when in front of him/her); the variable *Hidden humans* to "max" (i.e., let the robot be very prudent when entering rooms or turning corners in the corridor).

Table 3: Mapping of human and task classes to social motion parameters.

Classes	Safety	Visibility	Hidden Humans
Technical+Fragile	max	min	nominal
Technical+Average	min	nominal	nominal
Technical+Reliable	min	min	min
Interaction+Fragile	max	nominal	max
Interaction+Average	nominal	nominal	nominal
Interaction+Reliable	min	nominal	nominal
Social+Fragile	max	max	max
Social+Average	nominal	max	max
Social+Reliable	min	nominal	nominal

4 Experimental Evaluation

Experiments evaluate the capability of contextually adapting the navigation behaviors of the robot to the requirements of planned tasks. To this aim, the evaluation compares the results of two configurations: (i) *cohan* showing the behavior without the use of contextual information from the task planner; (iii) *cohan+platinum* showing the behavior of the proposed TAMP approach.

4.1 Simulation Experiments

We use simulation to evaluate the achieved *social autonomy* under different conditions. We show the velocity behaviors obtained with the two configurations of the control architecture for the modeled tasks.

Table 4: Classification of humans and tasks for the patrolling scenario.

Navigation Episode	Task	Human
Navigate corridor	Technical	Average
Enter patient's room	Social	Average
Monitor patient	Social	Fragile
Leave room	Social	Average

Patrolling scenario. Within the *patrolling* scenario, the goal is to navigate to rooms and monitor patients' state by approaching, taking into consideration their fragile conditions. Table 4 shows the parameters set for this scenario. The robot moves to each room to monitor the patient and the complete task consists of 13 navigation episodes. The velocity profiles of the robot and the time taken in each navigation episode are presented in Fig. 4. The left part of the figure shows the run in *cohan* configuration, while the right one shows the run in *cohan+platinum* configuration. The green line represents the start and the blue dashed line represents the end of each navigation phase. The subsequent figures follow the same conventions.

P1, P5, P9, and P13 are navigation episodes in the corridor, and the ones in between concern the entering, monitoring, and leaving patients' rooms. From the left part of Fig. 4 showing the *cohan* setting, it can be seen that the robot moves with a maximum velocity of around 0.7 m/s. However, this may not be ideal for monitoring patients with limited reactive abilities. Hence, the robot should be more careful around the patients which can be achieved by updating the parameters (*cohan+platinum*) as shown in the right part of Fig. 4.

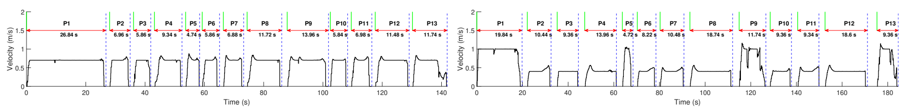


Fig. 4: Velocity profiles and times for each navigation phase in the patrolling scenario. Left: *cohan*, Right: *cohan+platinum*.

Since the corridor navigation episodes are technical, the robot could move with a larger velocity and use minimal human-aware capabilities for navigation. The total time taken for the robot to complete the task with dynamic parameters is 184.64 s while with constant parameters is 141.94 s. Results thus show that the integrated approach moves the robot more safely around the fragile humans in the environment.

Table 5: Classification of humans and tasks for the delivery scenario.

Navigation Episode	Task	Human
Navigate corridor	Technical	Average
Enter pharmacy	Interaction	Average
Approach pharmacist	Interaction	Reliable
Leave pharmacy	Interaction	Average
Enter patient's room	Social	Average
Deliver drugs	Interaction	Fragile
Leave room	Social	Average

Drug delivery scenario. Within the *drug delivery* scenario, a robot must take the prescribed drugs from the pharmacy and deliver them to a patient in a room. The complete set of parameters for this scenario is listed in Table. 5. Fig. 5 shows the velocity profiles and the execution times of each navigation episode.

The corridor navigation episodes in this scenario are P1, P5, and P9. The robot with dynamic parameters takes less time to traverse compared to the use of constant parameters. Episodes P2 to P4 are the interaction phases of the robot with the pharmacist who might have already interacted with the robot and could act reliably. The robot thus uses nominal human-aware parameters during these episodes.

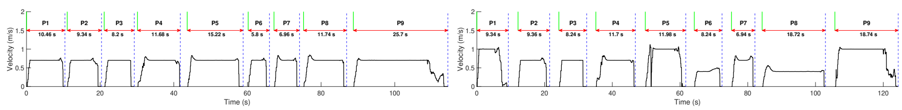


Fig. 5: Velocity profiles and times for each navigation phase in the drug delivery scenario. Left: *cohan*, Right: *cohan+platinum*.

The *cohan* setting takes 114.46 s while the *cohan+platinum* setting takes 124.46 s. However, the execution time for the *cohan+platinum* setting is lesser (103.26 s) than the *cohan* setting (105.1 s). Hence, the robot effectively spends less time navigating and delivering in the *cohan+platinum* configuration.

Table 6: Classification of tasks and humans for the emergency scenario.

Navigation Episode	Task	Human
Navigate corridor	Technical	Reliable
Enter patient's room	Technical	Average
Monitor patient	Technical	Fragile
Leave room	Interaction	Fragile

Emergency scenario. In the *emergency* scenario the robot has to rush to the patient to support an emergency call. For the setting, the list of parameters is updated as shown in Table. 6. Unlike the previous scenario, the robot moves faster in the *cohan+platinum* setting.

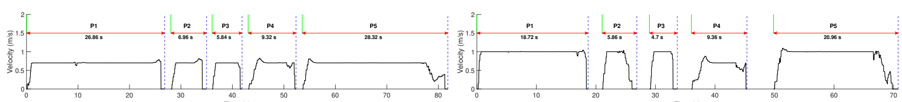
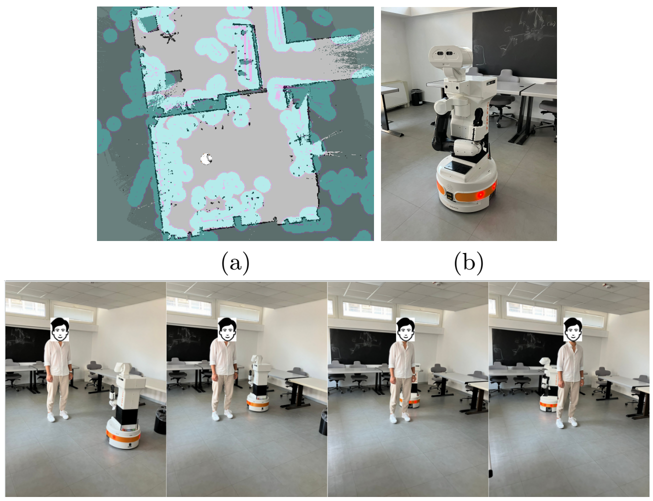


Fig. 6: Velocity profiles and times for each navigation phase in the emergency scenario. Left: *cohan*, Right: *cohan+platinum*

Fig. 6 shows that each episode in the *cohan+platinum* configuration is executed in less or almost the same time as the *cohan* setting. The total time for *cohan+platinum* setting is 70.80 s and it is 81.88 s in the *cohan* setting. This shows that the integrated approach takes less time to reach the patient in an emergency.Furthermore, the execution time for the *cohan* setting is 77.30 s which is larger than the overall planning time of the integrated approach. In the *cohan+platinum* setting, it is only 59.60 s. The benefits of the proposed TAMP approach is apparent from this scenario.

4.2 Laboratory Experiments

In addition to the simulated scenarios, we have replicated the assistive scenarios in the laboratory by deploying the TAMP approach on a PAL Tiago robot. Specifically, we have replicated the *drug delivery* scenario by mapping room, pharmacy, and corridor locations in the laboratory environment. Fig. 7(a) shows the layout of the laboratory. Locations of the simulated hospital scenarios have been sampled in the real map and used to implement robot navigation goals. Fig. 7(c) shows the robot crossing a human user while performing one of the navigation tasks of the plan. As in the simulation experiments, we have compared two configurations: (i) a configuration with static navigation parameters (*cohan*), and; (ii) a configuration with dynamic navigation parameters (*cohan+platinum*).



(a) (b)

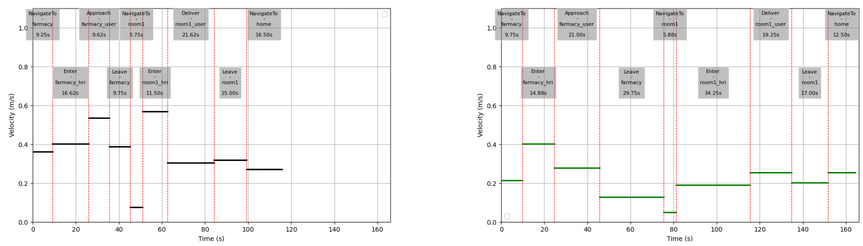
(c)

Fig. 7: Social navigation in the laboratory with the Tiago Robot.

Fig. 8 shows the average velocity recorded during the execution of the assistive task. Comparing the two configurations, it can be seen that *cohan+platinum* effectively adapts the robot's velocity to the changing social conditions of the navigation phases of the scenario. This is particularly evident comparing single navigation episodes. Consider, for example, the "Enter room1" episode, *cohan* has an average velocity close to 0.6 m/s while *cohan+platinum* has an average velocity close to 0.2 m/s. Similarly, a clear difference in the average velocity can be observed with the "Deliver room1" and "Leave room1" episodes. As a result indeed the execution of *cohan* is faster than the execution of *cohan+platinum*. The former execution takes less than 120 seconds while the latter takes more than 160 seconds. The behavior of the robot in the two configurations confirms the capability of the TAMP approach to tailor the navigation behavior dynamically.

5 Final Remarks and Future Works

We presented a novel integrated Task and Motion Planning approach to enhance the awareness of robot social navigation. We discussed the integration of task-level and geometric-level knowledge to tailor trajectory planning to contextual situations. We evaluated the proposed approach in a simulated assistive domain, considering three scenarios entailing different interaction requirements. We have



(a) (b)

Fig. 8: Recorded average velocity for each navigation episode of the delivery scenario: (a) *cohan* configuration; (b) *cohan+platinum* configuration.

also shown an initial laboratory evaluation with the approach deployed on a real social robot. Results show that the integrated approach allows the robot to adapt velocity and motion behaviors autonomously to various social situations. Future works are concerned with investigating ways to improve efficiency by reducing the overhead detected during the experiments. Also, we consider further integrating task and motion planners to improve social autonomy by dealing with exogenous events and contingencies during plan execution.

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